



Tutorial 2; Physical Chemistry

1. (a) The general expression for the electron tunneling (transmission) probability across a barrier is expressed as

$$T = \frac{|A'|^2}{|A|^2} = \frac{1}{1 + \frac{(e^{\kappa L} - e^{-\kappa L})^2}{16\varepsilon(1 - \varepsilon)}}$$

Find the transmission probability for high and wide barrier

- (b) An electron having total kinetic energy E of 4.50 eV approaches a rectangular energy barrier with $V = 5.00$ eV and $L = 950$ pm. Calculate probability of tunneling of this electron through the barrier (using the final expression derived in 1(a)).

2. Consider a scanning tunnelling microscope (STM). The tunnelling current is proportional to the tunnelling probability of electrons between the tip and the substrate surface. The separation between the tip and the substrate surface is $5\text{ \AA}$. The wavefunction between the tip and the substrate surface can be assumed to be

$$\Psi = Be^{-\kappa x}; \text{ where } \kappa = \frac{\{2m(V - E)\}^{1/2}}{\hbar}. \text{ Consider } (V - E) = 2.0 \text{ eV and use}$$

the formula for T derived for question 1. Find the factors by which the current would drop if the distances of separation were to be $6\text{ \AA}$, $8\text{ \AA}$ and $10\text{ \AA}$.

3. In a region of space, a particle with mass m and with zero energy has a time-independent one-dimensional wave function

$$\Psi = Axe^{-x^2/L^2}, \text{ where } A \text{ and } L \text{ are constants. Determine the expression for the potential energy } V(x) \text{ of the particle.}$$